

Bootstrapping Matrix SYK Models

Formulation, validation, and the road to large N

Project checkpoint

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Abstract

We develop a quantum-mechanical bootstrap for the purely fermionic “matrix SYK” models proposed by Chen [1] as the minimal chaotic supersymmetric systems exhibiting R -charge concentration. We (i) formulate the fermionic matrix bootstrap and validate it against Chen’s exactly-solvable single-matrix model $Q = \text{Tr}[\Psi^3]$; (ii) characterize the three-matrix model $Q = \sum_{i \leq j \leq k} \text{Tr}[\Psi_i \Psi_j \Psi_k]$ by exact diagonalization at $N = 2$, quantifying its dramatic R -charge concentration; (iii) reduce the single-matrix Hamiltonian to a purely single-trace quartic operator; and (iv) build a covariant single-trace bootstrap for the R -charge-resolved ground-state energy $E_0(N_\Psi)$, validated tight at $N = 2$. We identify the central obstruction — fortuitous states carry the *maximal* color Casimir and so admit no classical master field — and lay out the concrete, well-posed large- N target. All numerical claims are verified against exact diagonalization; code is itemized in Appendix A.

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1 Overview and program

Fortuity [2] is a proposed sharp feature of supersymmetric black-hole microstates: certain BPS states exist only at specific finite N and cannot be continued to $N = \infty$ while remaining BPS. Chen [1] argues that the minimal ingredients live in purely fermionic matrix quantum mechanics with a cubic supercharge — “matrix SYK” models — and proposes using the quantum-mechanical matrix bootstrap [3, 4] to study them. This document is a checkpoint of that program.

Two models organize everything:

- **Single matrix** (exactly solvable): one $U(N)$ -adjoint complex fermion matrix, $Q = \text{Tr}[\Psi^3]$. Non-chaotic; its BPS spectrum is fixed by representation theory. Our validation laboratory.
- **Three matrices** (the target): $Q = \sum_{1 \leq i \leq j \leq k \leq 3} \text{Tr}[\Psi_i \Psi_j \Psi_k]$. Chaotic and R -charge concentrated; genuinely interacting.

The logical arc: formulate the bootstrap (§3), validate it on the solvable model (§4), characterize the target by ED where possible (§5), then build the machinery that reaches beyond ED (§6). §7 summarizes status and complications.

2 The single-matrix model

2.1 Definition and algebra

A single $U(N)$ -adjoint complex fermion matrix $\Psi = (\Psi_{ij})$ (N^2 complex fermions) obeys

$$\{\Psi_{ij}, \bar{\Psi}_{kl}\} = \delta_{il}\delta_{jk}, \quad \{\Psi_{ij}, \Psi_{kl}\} = \{\bar{\Psi}_{ij}, \bar{\Psi}_{kl}\} = 0, \quad \bar{\Psi}_{ij} \equiv \Psi_{ji}^\dagger. \quad (1)$$

The $\mathcal{N} = 2$ supercharge and Hamiltonian are

$$Q = \text{Tr}[\Psi^3], \quad \bar{Q} = Q^\dagger = \text{Tr}[\bar{\Psi}^3], \quad H = \{Q, \bar{Q}\}. \quad (2)$$

Q is nilpotent, $Q^2 = 0$, so $H \geq 0$ with equality on BPS states. In an $\mathfrak{su}(N)$ basis, $Q = \frac{i}{\sqrt{2}} f^{abc} \psi^a \psi^b \psi^c$ and the trace mode decouples (doubling every degeneracy).

2.2 Symmetries and exact solution

The conserved quantities are the R -charge (fermion number) $N_\Psi = \text{Tr}[\Psi \bar{\Psi}]$ with $[N_\Psi, Q] = 3Q$, and the gauge $SU(N)$ generators J^a with Casimir $\hat{C}_2 = \sum_a J^a J^a$. A short computation gives the key identity

$$\boxed{H = 3N(N^2 - 1) - 9\hat{C}_2} \quad (3)$$

so the spectrum is fixed entirely by $SU(N)$ representation content — the model is non-chaotic. We verified (3) as an exact operator identity and reproduced the spectrum (Table 1).

Quantity	$N = 2$	$N = 3$
Hilbert space ($\mathfrak{su}(N)$ sector)	$2^3 = 8$	$2^8 = 256$
Spectrum ($E : \text{deg}$)	0(6), 18(2)	0(108), 18(80), 45(64), 72(4)
(E, C_2) pairs	(0, 2), (18, 0)	(0, 8), (18, 6), (45, 3), (72, 0)
$C_2^{\max} = N(N^2 - 1)/3$	2	8
$\dim \mathbf{r}_* = 3^{N(N-1)/2}$	3	27

Table 1: Exact-solution data for the single matrix, verified numerically. Energies lie exactly on $E = 3N(N^2 - 1) - 9C_2(\mathbf{r})$.

2.3 Fortuity

The BPS ($E = 0$) states fill the single “maximal representation” $\mathbf{r}_*$ — the staircase Young diagram with $N - 1$ rows and $2N - 2i$ boxes in row i , of maximal Casimir $C_2(\mathbf{r}_*) = N(N^2 - 1)/3$ and dimension $3^{N(N-1)/2}$. Because that diagram already saturates $N - 1$ rows, decreasing $N \rightarrow N - 1$ makes it illegal and the BPS states become null — the sharp N -sensitivity that defines fortuity. The BPS states occupy the R -charge window $\frac{N(N-1)}{2} \leq N_\Psi \leq \frac{N(N+1)}{2}$.

3 The fermionic matrix bootstrap: formulation

3.1 Master constraints

For a fixed energy eigenstate, expectation values $\langle O \rangle$ obey: positivity of the moment (Gram) matrix

$$M_{IJ} = \langle O_I^\dagger O_J \rangle \succeq 0, \quad (4)$$

the equations of motion (Heisenberg) $\langle [H, O] \rangle = 0$ and the stronger $\langle OH \rangle = E\langle O \rangle$, and — for the ground state — the zero-temperature thermal (“ground-state positivity”) condition $\langle O_I^\dagger [H, O_J] \rangle \succeq 0$. Extremizing $E = \langle H \rangle$ over all consistent functionals yields rigorous two-sided bounds.

3.2 Operators and Pauli truncation

Operators are words in the elementary fermions. Because each of the N^2 modes appears at most once (Pauli) and $\{\Psi, \bar{\Psi}\} = \delta\delta$, every word reduces to a canonical form and the independent-correlator set is *finite*: the algebra closes exactly at finite level, so the fermionic bootstrap is exact at finite level — unlike the continuum-moment bosonic case. The moment matrix block-diagonalizes by R -charge N_Ψ and by $SU(N)$ irrep.

3.3 The decisive subtlety: gauged vs. ungauged

Fortuity lives in the *ungauged, non-singlet* sector. In this model the singlet sector has $C_2 = 0$, hence sits at the *top* of the spectrum $E = 3N(N^2 - 1)$; a naive singlet-sector bootstrap would find the top, not the BPS states. One must retain non-singlet operators. Two equivalent formulations:

- (a) a fixed reference state, e.g. the highest-weight $|\lambda\rangle = \prod_{i>j} \Psi_{ij}|0\rangle$;
- (b) the multiplet-averaged (covariant) functional, $\langle O \rangle = \frac{1}{\dim} \text{Tr}_E[O]$, which is $SU(N)$ -invariant and generalizes cleanly.

3.4 The Casimir shortcut

Because (3) makes H affine in $\hat{C}_2$, for the single matrix *bounding the energy is bounding a Casimir*: $E_0(N_\Psi) = 3N(N^2 - 1) - 9C_2^{\max}(N_\Psi)$. This is what the dynamical bootstrap must reproduce, and precisely the shortcut that *disappears* for the three-matrix model.

4 Validation of the bootstrap on the single matrix

We implemented the truncated bootstrap as a linear functional $\phi(O) = \text{Tr}[\rho O]$ with ρ Hermitian but *not* positive (the most general functional; a returned value is a genuine bound, not ED). Results at $N = 2$ (formulation (a)):

Experiment	operators	result (exact)
$\min \phi(H)$, words only	level 2	-2.25 (genuine relaxation)
$\min \phi(H)$, $+\{Q, \bar{Q}\}$	level 2	≈ 0 ($= E_0$; SUSY sum-of-squares)
$\max \phi(H)$	level 2	18.0 ($= 3N(N^2 - 1)$)
$\max \phi(\hat{C}_2)$	level 1	2.0 ($= C_2^{\max}$)
feasibility of $ \lambda\rangle$	—	$M \succeq 0$, EOM $= 0$ (holds)

Table 2: $N = 2$ single-matrix bootstrap. Positivity plus the supercharge in the operator basis realizes $E \geq 0$; the maximal Casimir is recovered exactly.

R -charge-resolved energy. Imposing sharp charge and maximizing $\phi(\hat{C}_2)$ gives a rigorous lower bound on $E_0(N_\Psi)$. At level $2 + \{Q, \bar{Q}\}$ it is already tight and reproduces the BPS window exactly:

N_Ψ	0	1	2	3	4
E_0 exact	18	0	0	0	18
E_0 bootstrap	18	0	0	0	18

5 The three-matrix SYK model

5.1 Definition and verification

Three $U(N)$ -adjoint fermion matrices Ψ_i with $Q = \sum_{1 \leq i \leq j \leq k \leq 3} \text{Tr}[\Psi_i \Psi_j \Psi_k]$, $H = \{Q, \bar{Q}\}$. At $N = 2$ this is $pN^2 = 12$ complex fermions, $\dim = 2^{12} = 4096$. We verified it is a genuine $\mathcal{N} = 2$ theory: $Q^2 = 0$, $H = H^\dagger$, $[H, N_\Psi] = 0$, $[H, J^a] = 0$, $[N_\Psi, Q] = 3Q$.

5.2 R -charge concentration, quantified

Resolving the $E = 0$ states by N_Ψ (Table 3) makes concentration sharp: all 972 BPS states (24% of the Hilbert space) sit in just three sectors, $N_\Psi \in \{5, 6, 7\}$, with counts $243 : 486 : 243 = 3^5(1:2:1)$ centered at $n/2 = 6$. Contrast the single matrix, whose BPS states spread across the entire allowed window with binomial weights.

5.3 Interacting spectrum; chaos caveat

The full spectrum has only 168 distinct, *irrational* energies $(0.088, 0.226, 0.357, \dots)$ spanning $[0, 104.6]$ — qualitatively unlike the single matrix’s four integer levels. So the model is genuinely interact-

N_Ψ	≤ 4	5	6	7	≥ 8	total	$\langle N_\Psi \rangle$ (std)
#BPS	0	243	486	243	0	972	6.00 (0.71)

Table 3: BPS states of the $p = 3$, $N = 2$ matrix SYK model by R -charge. Exact, from ED.

ing. However, spectral chaos is *not* diagnosable at $N = 2$: the central $N_\Psi = 6$ sector (dim 924) has only ~ 50 distinct energies (gauge $SO(3) \times$ flavor $S_3 \times$ SUSY degeneracy), too few for a clean random-matrix test. Chaos is a large- N statement, inaccessible to ED — motivating the bootstrap.

6 Toward large N

6.1 The concentration question, made precise

Sparse ED of the single matrix at $N = 2, 3, 4$ (up to $\dim = 2^{16}$) gives $E_0(N_\Psi)$ exactly; the BPS window is $[\frac{N(N-1)}{2}, \frac{N(N+1)}{2}]$, i.e. $N + 1$ sectors — it *grows linearly with N* (Table 4). Crucially, at $N = 2$ the single-matrix window (3 sectors) equals the SYK window (3 sectors). The models are distinguished only by scaling:

Does the SYK BPS window stay $O(1)$ as N grows, while the single-matrix window grows as N ? This cannot be answered by ED (SYK at $N = 3$ is 2^{27}); it is exactly the large- N statement the bootstrap must supply.

N	BPS window	# sectors	dim
2	[1, 3]	3	16
3	[3, 6]	4	512
4	[6, 10]	5	65 536

Table 4: Single-matrix $E_0(N_\Psi)$ baseline (sparse ED). Window = $N + 1$ sectors, grows.

6.2 The single-trace Hamiltonian

A single-trace bootstrap never builds 2^{N^2} matrices. Matching to explicit H at $N = 2, 3$ (residual $< 10^{-11}$) gives the exact identity

$$H = \frac{9}{2} (\text{Tr}[\Psi\Psi\bar{\Psi}\bar{\Psi}] + \text{Tr}[\bar{\Psi}\bar{\Psi}\Psi\Psi]) - 9(N-1)^2 \quad (5)$$

with N -independent coefficient $9/2$ and a pure c-number remainder (the double-trace pieces of $\hat{C}_2$ collapse via $\text{Tr}[\Psi\bar{\Psi}] + \text{Tr}[\bar{\Psi}\Psi] = N^2\mathbf{1}$). The dynamics is thus *purely single-trace* — the favorable structure for large N .

6.3 The real obstruction

The fortuitous states carry the *maximal* color Casimir $C_2 \sim N^3$: they are as far from a gauge singlet as possible. Standard large- N technology (collective field / master field, and the published matrix bootstraps) assumes a near-singlet master field with single-trace factorization; that assumption fails for the fortuitous sector. This is not incidental — it *is* Chen’s point that fortuitous states have no classical large- N saddle. The correct tool is therefore a covariant, R -charge-resolved bootstrap that does not presuppose a master field.

6.4 The covariant single-trace bootstrap

We bootstrap $E_0(N_\Psi)$ using only gauge-invariant single-trace operators $\text{Tr}[\text{word}]$, with a general Hermitian functional (no factorization assumed), reduced to the touched operator span (SVD) so the number of SDP variables is $O(\text{tens})$, independent of 2^{N^2} .

Result 1 (it works, $N = 2$). With 26 variables the bootstrap reproduces $E_0(N_\Psi)$ *exactly* in every sector, including the lifted ones $(18, 0, 0, 0, 18)$. Gauge-invariant data alone bounds the non-singlet, maximal-Casimir ground state tightly: the master-field obstruction is not fatal.

Result 2 (where it strains, $N = 3$). At level 3 the bootstrap is tight *inside* the BPS window ($E_0 = 0$ recovered, easy by SUSY) but returns the trivial bound in the *lifted* sectors, where $E_0 = 72, 45, 18$. Proving $E_0 > 0$ is the hard direction and needs ground-state positivity and/or higher level — the standard convergence knobs. Adding them at $N = 3$ overruns our in-session sandbox (~ 45 s/call, ~ 4 GB), an environment limit, not a method failure (the same $N = 2$ run with ground-state positivity is tight everywhere).

7 Status, complications, next steps

Item	Status
Single-matrix model, exact solution, validation targets	done, verified
Fermionic matrix bootstrap formulation	done
Bootstrap validation on single matrix ($N = 2$; E_0 , $C_2^{\max}$, $E_0(N_\Psi)$)	done, tight
Three-matrix model build + $\mathcal{N} = 2$ verification	done
R -charge concentration quantified (ED, $N = 2$)	done
Single-matrix $E_0(N_\Psi)$ baseline $N = 2, 3, 4$	done
Single-trace form of H	done, verified
Covariant single-trace bootstrap ($N = 2$ tight; $N = 3$ window tight)	done
Lifted sectors at $N = 3, 4$ (ground-state positivity, higher level)	compute-limited
Abstract large- N single-trace algebra (no explicit matrices)	next
Three-matrix $E_0(N_\Psi)$ beyond $N = 2$	goal

Table 5: Program status.

Complications, honestly. (i) The non-singlet, maximal-Casimir nature of fortuitous states blocks the standard master-field bootstrap; the covariant formulation is the fix, validated at $N = 2$. (ii) Proving $E_0 > 0$ in lifted sectors (needed for concentration) is the hard direction and demands ground-state positivity + higher level, hence a real SDP solver (SDPB) and more compute than an in-session sandbox. (iii) Whether the covariant bootstrap *converges* tightly in the lifted sectors at $N = 3, 4$ is the immediate empirical question (Appendix B scripts).

Next steps. (1) Run the higher- N validation scripts (Appendix B) to (a) confirm the single-matrix window growth to $N = 5, 6$ and (b) attempt the three-matrix $E_0(N_\Psi)$ at $N = 3$ directly (the concentration test) where memory permits. (2) Port the covariant SDP to SDPB with sparse operator products; push $N = 3, 4$ lifted sectors with ground-state positivity and level 4–5. (3) Replace explicit matrices by the abstract single-trace algebra so N is a parameter; then bound $E_0(N_\Psi)$ for the three-matrix model at N far beyond ED and read off whether the window stays $O(1)$.

A Code inventory

File	Purpose
<code>mm_model.py</code>	single matrix, dense; $ \lambda\rangle$; ED
<code>mm_bootstrap.py</code>	truncated ρ -functional bootstrap ($N = 2$)
<code>sm_charge_profile.py</code>	R -charge-resolved $E_0(N_\Psi)$ bootstrap ($N = 2$)
<code>sm_charge_ED.py</code>	single-matrix $E_0(N_\Psi)$ baseline (sparse, $N \leq 4$)
<code>sm_trace_hamiltonian.py</code>	single-trace form of H
<code>sm_covariant_bootstrap.py</code>	covariant single-trace bootstrap
<code>msyk_model.py</code>	three-matrix SYK model (sparse)
<code>msyk_ed.py</code>	$N = 2$ ED: concentration, chaos

B Local validation scripts

See the accompanying `run_*.py` scripts, each labeled by model, for the higher- N validations to be run locally.

References

- [1] Y. Chen, *Fortuity with a Single Matrix*, arXiv:2511.00790.
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- [4] S. Laliberte, B. McPeak, *Bootstrapping supersymmetric (matrix) quantum mechanics*, arXiv:2510.01356.